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Multispectral state-space feature fusion: Bridging shared and cross-parametric interactions for object detection

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ABSTRACT

Modern multispectral feature fusion for object detection faces two critical limitations: (1) Excessive preference for local complementary features over cross-modal shared semantics adversely affects generalization performance; and (2) The trade-off between the receptive field size and computational complexity present critical bottlenecks for scalable feature modeling. Addressing these issues, a novel Multispectral State-Space Feature Fusion framework, dubbed MS2Fusion, is proposed based on the state space model (SSM), achieving efficient and effective fusion through a dual-path parametric interaction mechanism. More specifically, the first cross-parameter interaction branch inherits the advantage of cross-attention in mining complementary information with cross-modal hidden state decoding in SSM. The second shared-parameter branch explores cross-modal alignment with joint embedding to obtain cross-modal similar semantic features and structures through parameter sharing in SSM. Finally, these two paths are jointly optimized with SSM for fusing multispectral features in a unified framework, allowing our MS2Fusion to enjoy both functional complementarity and shared semantic space. Benefiting from the design of the dual-branch SSM, our approach simultaneously inherits the computational efficiency and the global receptive field, significantly improving the performance of multispectral object detection. Our experiments demonstrate that MS2Fusion consistently outperforms strong baselines: it improves mAP@0.5 by 3.3 % for object detection (FLIR), boosts mIoU by 2.8 % for RGB-T semantic segmentation (MFNet), and reduces MAE by 0.7 % for salient object detection (VT5000). Notably, without requiring task-specific modifications, MS2Fusion achieves new state-of-the-art results across multiple multispectral perception tasks, demonstrating its remarkable generalization capability. The source code is available at <https://github.com/61s61min/MS2Fusion.git>.

1. Introduction

Multispectral object detection has recently drawn increasing interest owing to its robust performance by fusing information from multiple spectral bands, such as RGB and thermal bands. RGB images usually offer high resolution, rich color and texture features, but suffer from sharply performance deterioration in complex scenarios such as low light, adverse weather or occlusions. In contrast, thermal images can effectively overcome these environmental limitations but exhibit obvious deficiencies in color and texture details. Current single-modal generic object detection methods struggle to overcome these aforementioned challenges. However, multispectral feature fusion paves a way

to provide a reliable object detection solution under such challenging conditions.

As shown in Fig. 1a, existing studies generally suggest that complementary features play a critical role when one modality is insufficient. For example, RGB images provide discriminative color and texture cues when thermal objects lack distinct contours, while thermal images offer thermal signatures when RGB imaging suffers from low illumination or occlusion. However, as demonstrated in Fig. 1b, scenarios where both modalities exhibit weak discriminative features (e.g., blurred textures in RGB and low contrast in thermal) cannot be resolved by complementary information alone. In such cases, shared features, such as cross-modal consistent shapes and structural patterns, become essential, as they cap-

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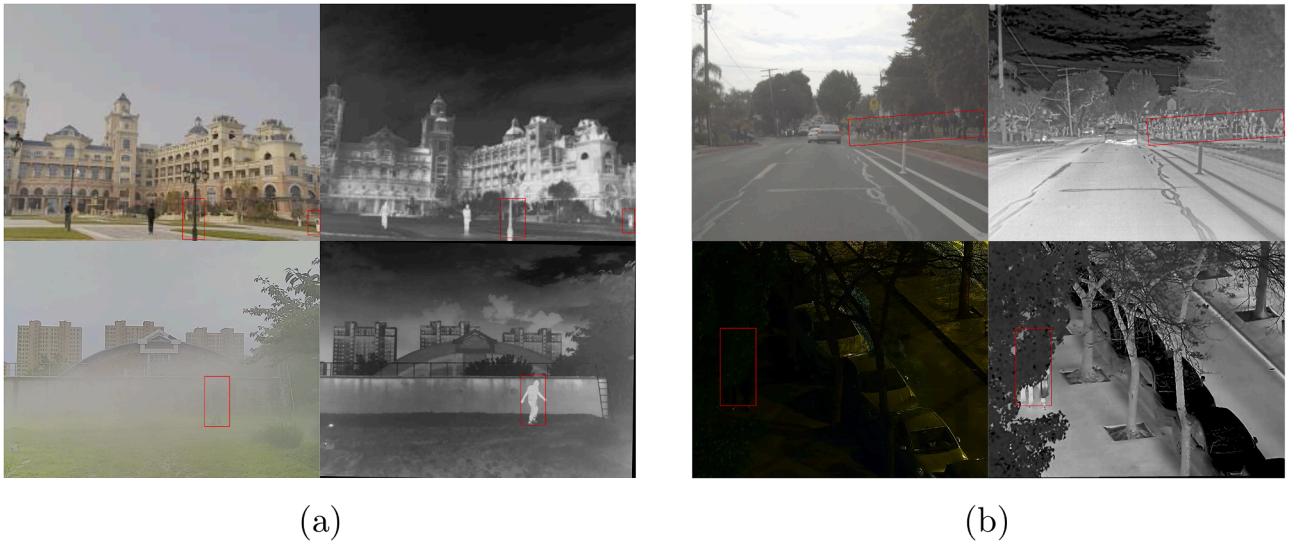


Fig. 1. The pros and cons of RGB (left) and thermal (right) images. (a) Both modalities provide complementary information, and their fusion enables more robust object detection; (b) Dual-modal shared features become crucial, since neither modality stands out distinctly. (e.g., modality-specific characteristics such as texture and thermal radiation are blurred, and cross-modal consistent features like object contours and structures are helpful for detection).

ture modality-invariant representations for reliable detection. Thus, we speculate that a robust multispectral object detection framework should dynamically leverage both complementary and shared features to handle diverse real-world challenges.

Previous studies [1–7] have predominantly focused on complementary feature learning across modalities, often overlooking the exploration of shared feature representations or inherent structural similarities between them. Moreover, existing approaches typically employ a single fusion strategy to directly combine multi-modal inputs, neglecting the potential benefits of adaptive or hierarchical fusion mechanisms. These methods do not fully explore or exploit cross-modal shared features, ignoring the potential effect of enhancing the performance of single-modal features. This fusion paradigm often suppresses weaker yet discriminative features during cross-modal integration, leading to significant information loss. For multispectral object detection, shared feature representation plays a pivotal role in multi-modal fusion. Not only does it mitigate cross-modal discrepancies, but it also augments single-modal features, thereby substantially improving feature expressiveness and detection robustness in challenging environments.

In addition, most of the mainstream methods leverage CNN [1] or Transformer [2,8,9] for feature fusion. Albeit effective, existing CNN-based methods often struggle with capturing broader contextual information across modalities due to their limited receptive fields. On the other hand, despite excellence at modeling global dependencies, the Transformer-based approaches may degrade with longer input sequences, leading to performance degradation and higher computational costs as model complexity. These factors restrict their practicality in resource-constrained environments, underscoring the need for more efficient fusion strategies in multispectral object detection.

Despite recent advances in multispectral object detection, existing methods predominantly focus on learning complementary features across modalities while neglecting the exploration of shared feature representations and inherent structural similarities. Most approaches adopt a single, direct fusion strategy, which often suppresses discriminative yet weaker features and leads to significant information loss. Moreover, while CNN-based methods struggle with limited receptive fields for capturing cross-modal context, Transformer-based techniques suffer from computational inefficiency and performance degradation with long sequences. In speech recognition [10] and image compression [11], the efficiency of feature representation directly determines system performance - a principle equally applicable to multispectral detection. These

limitations highlight the need for a more adaptive and efficient fusion paradigm that not only bridges modal discrepancies but also enhances single-modal features through shared representations, ultimately improving robustness in challenging environments.

Recent advances in sequence modeling show SSM-based methods excel by compressing features into compact hidden states, enabling efficient inference with constant-time full-sequence processing. Mamba [12] enhances this with selective state spaces, dynamically retaining task-relevant features. Vision Mamba (Vim) [13] further proves its effectiveness for visual tasks, boosting both efficiency and performance.

Inspired by this, we propose MS2Fusion, a novel framework that simultaneously leverages complementary and shared features across modalities while overcoming limitations of CNN and Transformer in multispectral feature fusion. Fig. 2 reveals that Transformer-based methods (a) and existing Mamba solutions (b) fail to exploit cross-modal shared features. In contrast, our method (c) explicitly models both complementary and shared cross-modal interactions through three key components:

- **Cross-Parametric State Space Model (CP-SSM):** Facilitates implicit feature complementarity by exchanging output matrices between modality-specific state spaces, enabling cross-modal feature enrichment while preserving modality-specific characteristics.
- **Shared-Parametric State Space Model (SP-SSM):** Learns a unified feature space through parameter sharing, aligning heterogeneous modality distributions to extract discriminative shared representations that enhance single-modality features.
- **Feature Fusion State Space Model (FF-SSM):** Introduces a bidirectional input scheme to expand the Effective Receptive Field (ERF) of state spaces, mitigating feature attenuation while enabling adaptive fusion of cross-modal information.

As demonstrated in Fig. 3, our ERF analysis reveals that MS2Fusion achieves superior spatial coverage compared to both CNN and Transformer, successfully integrating local details with global context. This capability addresses the fundamental constraint of standard CNN in modeling long-range dependencies while maintaining computational efficiency. Table 1 also provides our component replacement experiments on the MS2Fusion module in the YoloV5 framework on FLIR dataset (Fig. 3), which yield three key findings: (1) Compared to Transformer, MS2Fusion achieves a 66.6% reduction in FLOPs and 70.4% fewer parameters while delivering better detection accuracy

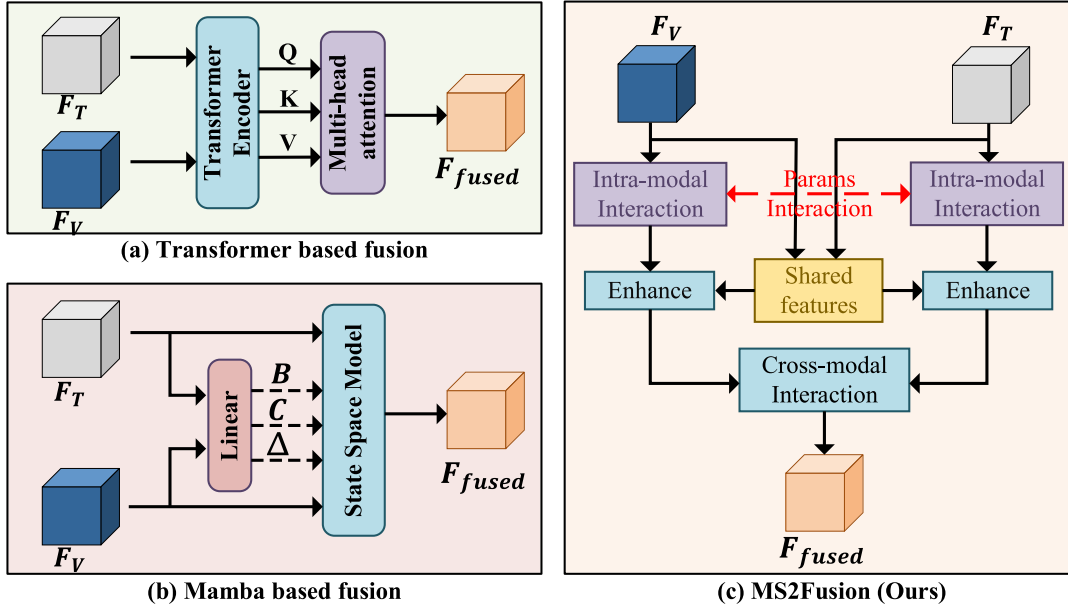


Fig. 2. Comparing Transformer-based fusion (a), Mamba-based fusion (b) and our proposed MS2Fusion (c), with F_T and F_V as the input thermal and RGB image features, respectively. In Transformer-based method (a), F_T and F_V are fused through the multi-head attention mechanism, effectively integrating complementary information and enhancing performance across scenarios. The traditional Mamba approach (b) directly mixes dual-modal features to generate B , C , and Δ parameters for SSM-based feature interaction, which may lead to modal misalignment and feature redundancy. In contrast, our method (c) first performs intra-modal feature interaction and then extracts cross-modal shared features, achieving better modal alignment and fusion, thereby providing more robust and unified feature representations.

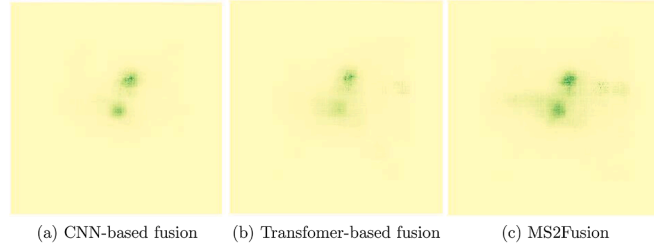


Fig. 3. Effective receptive field visualizations comparing CNN-based fusion method (a), Transformer-based fusion method (b), and the proposed MS2Fusion (c) method. Quantitative analysis demonstrates that MS2Fusion achieves significantly broader receptive field coverage compared to the others.

Table 1

Comparison of different multispectral feature fusion methods (YoloV5 framework, FLIR dataset).

	GFLOPs	Params (M)	FPS	mAP@0.5	mAP@0.75	mAP
CNN	190.3	159.7	30.8	74.3	23.8	32.5
Transformer	421.9	440.6	19.2	75.0	24.1	33.4
MS2Fusion	140.8	130.3	29.7	83.3	33.0	40.3

(+ 8.3 mAP50 points); (2) When benchmarked against CNN baseline, it maintains 26.0% lower computational costs and 18.4% parameter reduction, while achieving significant improvements of +9.0 mAP50 points; (3) Across all complexity-accuracy trade-off metrics, MS2Fusion consistently outperforms both baseline models. Notably, our approach achieves processing speed comparable to CNN (with only a 2.1 frames difference) while being significantly faster than Transformer (by 10.5 frames). These systematic experiments provide conclusive evidence that MS2Fusion successfully breaks the traditional efficiency-accuracy trade-off, establishing new state-of-the-art performance in multispectral object detection.

Our contributions are as follows:

- ♦ To address the limitations of existing methods that either overlook cross-modal interactions or overly rely on complementary features, MS2Fusion introduces a novel state-space framework. Its CP-SSM captures implicit feature complementarity while SP-SSM enhances shared representations, achieving an optimal balance between modality-specific and cross-modal learning.
- ♦ Motivated by the need of combining global context modeling with computational efficiency, our FF-SSM mechanism uniquely achieves Transformer-level receptive fields while maintaining CNN-like efficiency. This breakthrough resolves the longstanding trade-off between performance and computational cost in multimodal systems.
- ♦ MS2Fusion architecture offers flexible multimodal integration, supporting various detection frameworks and backbones. It seamlessly processes RGB, thermal, and other modalities, enabling broad application across different vision tasks without architectural changes.
- ♦ The MS2Fusion achieves state-of-the-art performance on benchmark datasets, including RGB-T object detection, semantic segmentation, and salient object detection, validating its effectiveness for multi-spectral image perception.

The rest of this paper is organized as follows. [Section 2](#) reviews related research on multi-spectral object detection and Mamba. [Section 3](#) describes the proposed method. [Section 4](#) presents experimental results

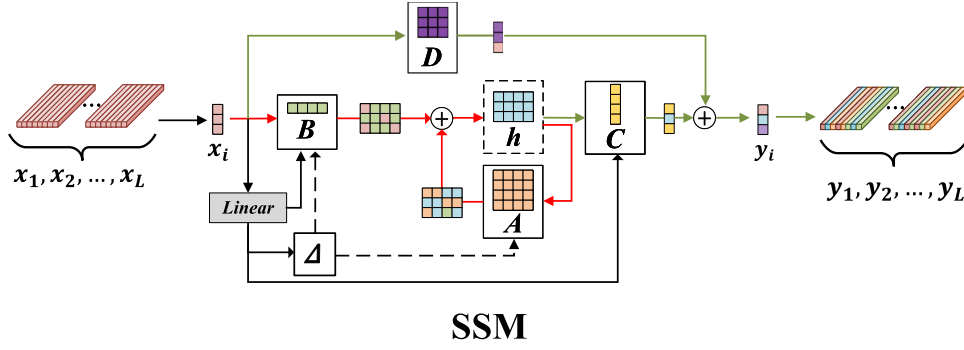


Fig. 4. Details of SSM, where the red, green and black lines correspond to the equations of Equation (2), respectively. (For an $L \times d$ dimensional input $\{x_1, x_2, \dots, x_L\}$.) The input sequence is linearly projected to generate parameters for the state equation, followed by recursive computation via matrices A , B , and C , with an optional skip connection D . The output sequence retains the same dimensionality as the input.

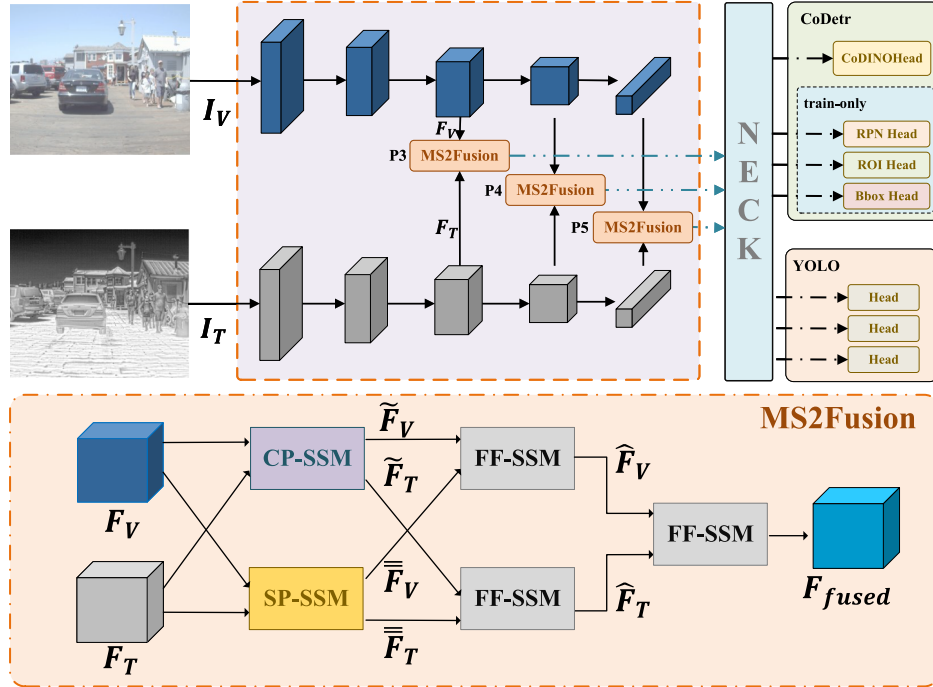


Fig. 5. Overview of the model architecture. It consists of three main stages: (1) feature extraction with two backbone networks; (2) cross-modal feature fusion of $P3$, $P4$ and $P5$ with MS2Fusion module; (3) the detection results are generated through the Neck and Head layers. In our experiments, two distinct detection heads (CoDettr and YOLOv5) are evaluated independently. The MS2Fusion module employs a dual-branch architecture to process the features of two modalities, F_V and F_T . CP-SSM learns cross-modal complementary features, while SP-SSM extracts shared features. Finally, FF-SSM performs single branch feature enhancement and cross-modal fusion, outputting the fused feature F_{fused} .

and analysis. Finally, we summarize the main points of the paper in Section 5.

2. Related works

2.1. Object detection

In object detection, RGB images are typically used for unimodal detection. There are two main approaches: two-stage detectors and one-stage detectors. Two-stage detectors (e.g., R-CNN [14], Fast R-CNN [15], Faster R-CNN [16], Mask R-CNN [17]) first generate region proposals and then perform classification and bounding box regression. Single-stage detectors (e.g., YOLO [18], SSD [19], RetinaNet [20]) perform object detection directly on the image without generating region proposals, resulting in faster detection speeds.

Anchor-based methods in object detection utilize predefined anchor points, each representing specific sizes and aspect ratios, to detect ob-

jects through regression adjustments. In contrast, anchor-free methods such as CornerNet [21], FCOS [22], and CenterNet [23] predict object boundaries or centers directly without anchors, simplifying the detection process with improved efficiency and often higher accuracy.

More recently, DETR (DEtection TRansformer) [24] introduced a fully end-to-end approach by leveraging Transformer to eliminate the need for hand-designed components like anchors or non-maximum suppression (NMS). DETR treats object detection as a set prediction problem, using bipartite matching to assign predictions to ground truth objects. While achieving competitive accuracy, its computational cost and slow convergence remain challenges, prompting follow-up improvements like Deformable DETR [25].

To fully validate the effectiveness of the proposed method, we selected the anchor-based YOLOv5 detection framework and the Transformer-based CoDettr detection framework for comparative experiments. The experimental results show that the proposed method offers the following advantages: 1) linear time complexity, ensuring high com-

putational efficiency; 2) a global receptive field characteristic, enabling the capture of richer contextual information.

2.2. Multispectral object detection

Recent advances in multispectral object detection have made significant progress in addressing two core challenges: cross-modal feature fusion and environmental adaptability. Early approaches focused on balanced feature integration, with methods like the Cyclic Fusion Module [26] explicitly modeling both complementarity and consistency between modalities. Subsequent works introduced more sophisticated attention mechanisms to dynamically weight features, such as Guided Attention Feature Fusion (GAFF) [27], which employed adaptive intra-modal and cross-modal attention to enhance fusion performance.

The emergence of Transformer-based architectures has further advanced the field by capturing long-range dependencies between modalities. The Cross-Modal Fusion Transformer (CFT) [2] demonstrated the effectiveness of self-attention for global contextual fusion, while CMX [28] improved generalization through feature rectification modules. Recent variants like ICAFusion [8] and INSANet [29] have optimized efficiency and flexibility, using parameter-shared Transformer and dedicated spectral attention blocks, respectively.

Several innovative approaches have pushed the boundaries of multispectral detection by addressing modality-specific challenges. DAMSDet [30] tackles modality misalignment and dynamic complementary characteristics through deformable cross-attention, while MS-DETR [31] introduces reference-constrained fusion to improve RGB-thermal alignment. Lightweight designs have also gained attention, such as the CPCF module [32], which combines channel-wise and patch-wise cross-attention for efficient fusion. Meanwhile, TFDet [33] employs a fusion-refinement paradigm with adaptive receptive fields to suppress false positives. Most recently, MMFN [34] proposed a comprehensive hierarchical fusion framework, integrating local, global, and channel-level interactions for robust multispectral detection.

Our method employs a novel SSM for feature fusion, which is different from the former methods using CNN or Transformer for feature fusion.

2.3. Mamba

Mamba [12] is a selectively structured state-space model designed for effective long sequence modeling tasks. It overcomes the limitations of CNN by incorporating global perceptual fields and dynamic weighting, achieving advanced modeling capabilities akin to Transformer but without their typical quadratic complexity. Building on Mamba's foundation, VMamba [35] is a visual state-space model that introduces the Cross Scan Module (CSM) to enhance scanning efficiency across dimensions, surpassing both CNN and ViTs in performance for computer vision tasks. Meanwhile, VM-UNet [36] integrates Mamba into the UNet framework for medical segmentation, leveraging visual state-space blocks to capture extensive contextual information. Additionally, Mamba is applied to multimodal semantic segmentation [37], enhancing global receptive field coverage with linear complexity through a Siamese encoder and innovative fusion mechanisms.

Inspired by these advancements, we propose MS2Fusion based on Mamba dynamic state space. This approach effectively harnesses shared features and complementarities between modalities, enhancing object detection accuracy while reducing model complexity.

3. Methods

3.1. State space model (SSM)

SSM [12] combines the advantages of RNNs and CNNs for efficient handling of long dependencies. It transforms the input sequence $x(t)$ into an intermediate state $h(t)$ through a state equation, then generates the

output $y(t)$ via an output equation. SSM is typically represented as a linear ODE:

$$\begin{aligned} h'(t) &= \mathbf{A} \cdot h(t) + \mathbf{B} \cdot x(t) \\ y(t) &= \mathbf{C} \cdot h(t) + \mathbf{D} \cdot x(t) \end{aligned} \quad (1)$$

To adapt SSM for deep learning, it is discretized as:

$$\begin{aligned} h_k &= \bar{\mathbf{A}} \cdot h_{k-1} + \bar{\mathbf{B}} \cdot x_k \\ y_k &= \bar{\mathbf{C}} \cdot h_k + \mathbf{D} \cdot x_k \\ \bar{\mathbf{A}} &= \exp^{\Delta \mathbf{A}} \\ \bar{\mathbf{B}} &= (\exp^{\Delta \mathbf{A}} - \mathbf{I}) / \Delta \mathbf{A} \\ \bar{\mathbf{C}} &= \mathbf{C} \end{aligned} \quad (2)$$

After discretization [38], SSM uses convolution operations for parallel computation. To overcome the issue of fixed parameters, SSM introduces dynamic adjustments, allowing the model to flexibly handle complex data and improve long-sequence modeling.

As shown in Fig. 4, the SSM processes the input sequence $\{x_1, x_2, \dots, x_L\} \in \mathbb{R}^{L \times d}$ through a structured transformation composed of three key operations. First, the input undergoes a linear projection to generate parameters for the state space equation. This is followed by a recursive state update governed by four fundamental matrices: the current input is initially processed by matrix $\mathbf{B}$, which maps the input to the state space, and then added to the hidden state updated by matrix $\mathbf{A}$, producing a new hidden state. For the output, the new hidden state is remapped to the output via matrix $\mathbf{C}$. Additionally, $\mathbf{D}$ provides an optional skip connection that bypasses the state transformation, directly adding the input to the output, which helps mitigate gradient vanishing during training. This recurrent computation generates intermediate outputs $\{y_1, y_2, \dots, y_L\} \in \mathbb{R}^{L \times d}$ [38], where each $y_i \in \mathbb{R}^{1 \times d}$ is derived sequentially.

3.2. The proposed model architecture

As shown in the upper part of Fig. 5, the proposed framework is structured into three main stages. Initially, the backbone network extracts features independently from RGB and thermal images. Subsequently, cross-modal MS2Fusion feature fusion integrates features from different stages. Finally, object localization and regression are performed using the detection head to derive the final detection outcomes. During the feature fusion stage, we selectively merge features from three distinct levels. P3 layer captures detailed surface information, while P5 encapsulates higher-level semantic feature. By separately fusing shallow and deep features, our approach effectively concentrates on detailed information critical for cross-modal fusion, which is formulated in Equation (3):

$$\begin{aligned} F_{fused}^i &= \phi_{MS2Fusion}(F_V^i, F_T^i) \\ [bbox, cls] &= \Phi_{head}(\Psi_{neck}(F_{fused}^3, F_{fused}^4, F_{fused}^5)) \end{aligned} \quad (3)$$

where F_V^i, F_T^i denote the feature maps in layer i extracted from the input RGB and thermal image with two backbones, and $\phi_{MS2Fusion}$ is the proposed state-space fusion module, F_{fused}^i denotes the fused feature. FPN [39] and PANet [40] are usually used as $\text{neck}(\Psi_{neck})$ to aggregate multi-scale features, while the detection head (Φ_{head}) is used for bounding box classification and regression. In our experiments, two different detection heads (YOLOv5 and CoDet) are evaluated separately.

3.3. Multispectral state space feature fusion (MS2fusion)

As shown in the lower part of Fig. 5, the MS2Fusion module consists of three core components: CP-SSM, SP-SSM, and FF-SSM. CP-SSM module achieves global feature integration of RGB and thermal modalities through a dynamic parameter interaction mechanism, enhancing cross-modal contextual awareness while preserving modality-specific

characteristics. This module innovatively employs an implicit parameter crossover strategy to effectively mine and reinforce complementary features between the two modalities. CP-SSM allows adaptive parameter modulation based on inter-modal relationships, enabling finer-grained complementary feature extraction. In contrast, the SP-SSM is designed to learn shared, modality-invariant representations that are consistent across both RGB and thermal inputs. All parameters in SP-SSM are constrained to be identical between modalities, forcing the model to discard modality-specific noise and to focus on common semantic information, such as object contours, shapes, and scene structure.

The key distinction between CP-SSM and SP-SSM lies in their parameter handling strategies and functional objectives. The CP-SSM employs dynamic parameter interaction mechanisms where modality-specific parameters are selectively exchanged or combined to capture complementary cross-modal dependencies, while the SP-SSM strictly maintains identical parameters across modalities to enforce feature consistency in shared spaces.

Finally, the FF-SSM modules further integrate these learned features in a hierarchical manner, adaptively combining complementary and shared information to achieve more effective cross-modal fusion.

The innovation of MS2Fusion lies in its systematic utilization of the dual characteristics of multispectral data: the CP-SSM explores cross-modal complementary information, while the SP-SSM strengthens modality-invariant shared features, ultimately achieving optimal feature fusion through the FF-SSM. This dual-path design explicitly decouples the learning of complementary and shared features: CP-SSM's parameters are optimized for cross-modal differentiation, whereas SP-SSM's parameters are optimized for cross-modal unification. By simultaneously focusing on modality-specific complementary features and modality-invariant shared features, the framework establishes a comprehensive cross-modal feature collaboration mechanism that significantly enhances the representational capability of multispectral data.

3.3.1. CP-SSM Module

As illustrated in Fig. 6, there are two branches in the CP-SSM module, RGB feature (top) and thermal feature branch (bottom). In this section, we only describe the RGB feature branch for clarity. The procedure of the thermal feature branch is identical.

Given $F_V \in \mathbb{R}^{d \times H \times W}$ from the RGB feature map, it is unfolded into a sequence $(x_1, x_2, x_3, \dots, x_L) \in \mathbb{R}^{d \times L}$, where d, H, W denote the channel, height and width of the feature map, respectively and $L = H \times W$. Following the Mamba mechanism in Section 3.1, the unfolded sequences are passed through a linear layer to obtain B, C, Δ , where $B \in \mathbb{R}^{L \times d \times d'}$, $C \in \mathbb{R}^{L \times d \times d'}$, $\Delta \in \mathbb{R}^{L \times d}$, and d' is the dimension of the hidden state. In the CP-SSM module, we innovatively designed a cross-parameter interaction mechanism that achieves implicit feature fusion through real-time interaction between the dual-modal state space projection matrices C_V and C_T . Specifically, this mechanism establishes bidirectional feature enhancement channels: in the RGB modality branch, the response patterns of thermal features are selectively fused via the exchanged C_T matrix, while the thermal branch enhances its feature discriminability by incorporating the semantic prior information encoded in the C_V matrix. This cross-parameter interaction strategy possesses two key characteristics: (1) it maintains the independence of modality-specific features to avoid confusion, and (2) it constructs an implicit attention mechanism at the state space dimension, enabling the complementary feature fusion process to be self-adaptive. Finally, after the reverse operation of folding the sequence $(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3, \dots, \tilde{x}_L)$, the output $\tilde{F} \in \mathbb{R}^{d \times H \times W}$ for each branch is obtained in Equation (4):

$$\begin{aligned} \tilde{F}_i &= \phi_{SSM}^i \left(F_{flip}^i(F_i), \Delta_i, B_i, C_i \right) \\ [\Delta_i, B_i, C_i] &= L_{Linear}(F_i), i \in \{V, T\} \end{aligned} \quad (4)$$

where F_{flip}^i indicates that the feature map is expanded in a certain way into a sequence, B_i, C_i, Δ_i obtained from the input sequence through the fully connected layer, and ϕ_{SSM}^i denotes the SSM in Fig. 4.

It is worth noting that, inspired by the VSSBlock in VMamba [35], we have explored three unfolding methods: unfold by rows, unfold by columns and unfold with both. We will provide detailed ablation studies in Table 13.

3.3.2. SP-SSM Module

The SP-SSM module constructs a hierarchical feature-sharing architecture, achieving cross-modal representation alignment through parameter sharing and feature reconstruction. As shown in Fig. 7, the pipeline of this module consists of two meticulously designed stages:

In the parameter-sharing stage, the module employs a coarse-grained additive feature fusion method to preliminarily integrate the RGB and thermal features. Then, three sets of critical shared parameters are dynamically generated through a linear network. These parameters not only encode the common feature patterns of the dual modalities but also retain the modality-specific adjustment capabilities.

In the feature reconstruction stage, these shared parameters are injected into the SSM of both modalities. This design creates a dual-stream coupled architecture: on one hand, the shared parameters constrain the evolution trajectories of the two modalities in the state space, driving heterogeneous features to converge into a shared feature space; on the other hand, each modality retains independent initialization states, ensuring the integrity of modality-specific information. Through this shared-parameter computation paradigm, the module can extract deep shared features that are invariant to factors such as lighting conditions and environmental interference. The specific process is formulated in Equation (5):

$$\begin{aligned} \bar{F}_i &= \phi_{SSM}(F_i, \Delta_s, B_s, C_s) \\ [\Delta_s, B_s, C_s] &= L_{Linear}(F_V \oplus F_T) \end{aligned} \quad (5)$$

where $\bar{F}_i$ ($i \in \{V, T\}$) are the shared features and ϕ_{SSM} is the SSM. Δ_s, B_s, C_s are obtained from the weighted fusion features through a fully connected layer.

3.3.3. FF-SSM Module

Although the CP-SSM module can capture implicit complementary relationships between modalities, it still has limitations at the feature fusion level. To address Mamba's inherent feature forgetting while preserving its computational efficiency, we propose a bidirectional architecture combining complementary forward (F_1, F_2) and reverse (F_2, F_1) processing paths, which is shown in Fig. 8. This innovative design systematically compensates for unidirectional modeling limitations through direction-adaptive parameter generation, where the reverse path specifically aims to capture potential feature loss in the forward pass. Unlike traditional Transformers that suffer from information loss due to chunking operations, our approach not only maintains Mamba's original sequential dependency and linear complexity advantages but also significantly enhances feature retention through reverse-sequence compensation. The dual-path mechanism enables more comprehensive capture of long-range cross-modal dependencies while dynamically mitigating feature forgetting, achieving robust sequential modeling without sacrificing efficiency. Specifically, given the feature maps F_1 and F_2 output by CP-SSM, this module employs a bidirectional heterogeneous sequence construction strategy: concatenating the features in different orders $(F_1, F_2$ and $F_2, F_1)$. This design not only expands the model's receptive field but also enhances the diversity of feature representation. In implementation, the features are first unfolded and concatenated for the two directional sequences, followed by generating corresponding state space model parameters $(B_1, C_1, \Delta_1$ and $B_2, C_2, \Delta_2)$ through linear transformation layers. Finally, by fusing the output features $(F_{12}$ and $F_{21})$ from the bidirectional SSM paths, the module achieves thorough cross-modal feature interaction and adaptive fusion, significantly improving detection accuracy. The FF-SSM module effectively preserves the detailed information within both modalities, ensuring richer and

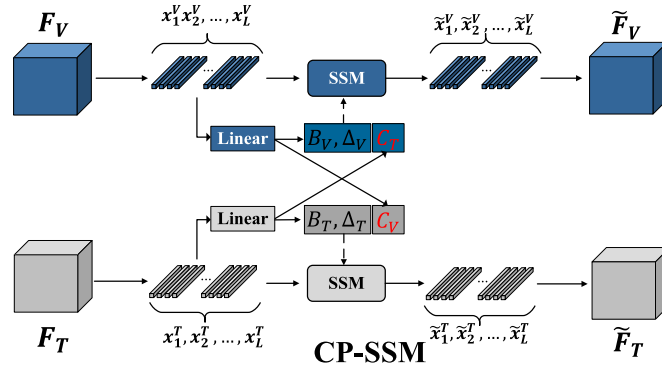


Fig. 6. The details of the CP-SSM module. The feature maps (F_V, F_T) are first reshaped into a sequence (x_i^V, x_i^T) by row and column scanning, and generated B, C, Δ through a Linear layer. Secondly, we perform a cross-modal complementary features interaction by exchanging the C of the two branches. Finally, the cross-modal complementary feature interaction is conducted by the SSM module to generate the output ($\tilde{F}_V, \tilde{F}_T$).

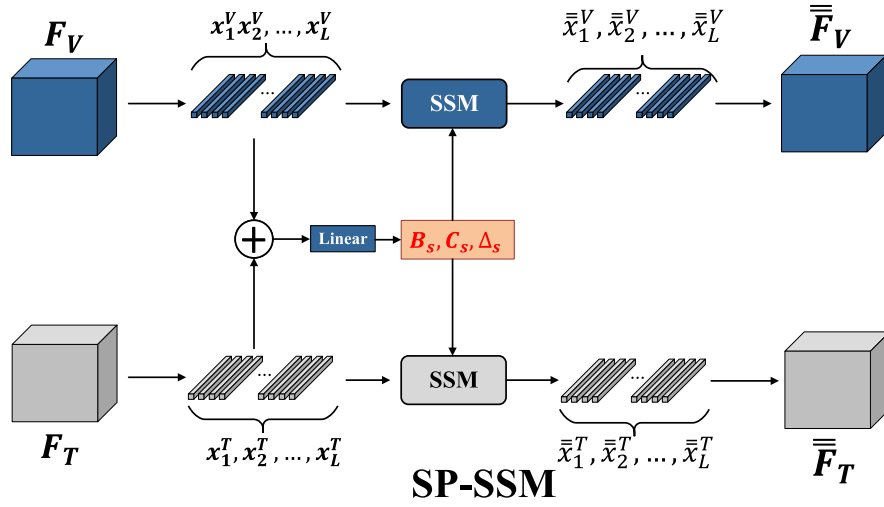


Fig. 7. The details of the SP-SSM module. The SP-SSM module extracts shared features in two modalities (RGB and thermal) from the SSM with shared parameters. The input feature F_V and F_T are combined to generate parameter B_s, C_s, Δ_s , while the output feature $\bar{\bar{F}}_V$ and $\bar{\bar{F}}_T$ are reconstructed by the two SSMs.

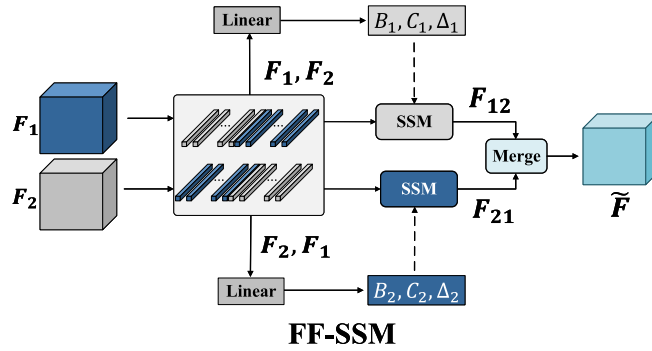


Fig. 8. The details of the FF-SSM module. It fuses features by combining F_1 and F_2 in two different orders. In the top path, the input features are combined in 1–2 order (e.g., F_1, F_2) to form the splice feature to generate B_1, C_1, Δ_1 by a Linear layer, and then cross feature interactions are performed via SSM. Finally, the F_{12} and F_{21} are merged to generate the fused feature map $\tilde{F}$.

more precise feature representation, which is formulated in Equation (6):

$$\begin{aligned}
 \hat{F} &= \psi_{Merge}(F_{12}, F_{21}) \\
 F_{12} &= \varphi_{SSM}([F_1, F_2], \Delta_1, B_1, C_1) \\
 F_{21} &= \varphi_{SSM}([F_2, F_1], \Delta_2, B_2, C_2) \\
 [\Delta_1, B_1, C_1] &= L_{Linear}^1(F_1, F_2) \\
 [\Delta_2, B_2, C_2] &= L_{Linear}^2(F_2, F_1)
 \end{aligned} \tag{6}$$

where φ_{SSM} denotes the SSM in Fig. 4, F_1, F_2 denotes the input feature maps from RGB and thermal modalities, ψ_{Merge} denotes the merging of the outputs of two sequences with different connection orders after the SSM. The Δ_i, B_i, C_i are the corresponding parameters in SSM.

3.4. Loss function

The fused features produced by MS2Fusion are fed into the neck and detection head, with the entire pipeline being optimized end-to-end. In this paper, we have evaluated MS2Fusion in both YOLOv5 and Co-Detr based detection frameworks.

In the YOLO framework, the total loss function can be formulated as Equation (7):

$$\mathcal{L}_{yolo} = \lambda_{bbox} \cdot \mathcal{L}_{bbox} + \lambda_{obj} \cdot \mathcal{L}_{obj} + \lambda_{cls} \cdot \mathcal{L}_{cls} \quad (7)$$

where $\mathcal{L}_{bbox}$, $\mathcal{L}_{obj}$ and $\mathcal{L}_{cls}$ are the localization loss, confidence loss and classification loss, respectively. The hyperparameters λ_{bbox} , λ_{obj} and λ_{cls} adjust the loss weights, which are kept as the default settings of the baseline.

The CoDetr framework employs a multi-task loss function consisting of four components: a primary CoDINOHead [41] and three auxiliary detection heads (RPN head, ROI head, and Bbox head). The total loss can be formulated in Equation (8):

$$\begin{aligned} \mathcal{L}_{CoDetr} = & \lambda_{primary} \cdot (\mathcal{L}_{QFL} + \mathcal{L}_{L1} + \mathcal{L}_{GloU}) + \lambda_{RPN} \cdot (\mathcal{L}_{CE} + \mathcal{L}_{L1}) \\ & + \lambda_{ROI} \cdot (\mathcal{L}_{CE} + \mathcal{L}_{GloU}) + \lambda_{Bbox} \cdot (\mathcal{L}_{Focal} + \mathcal{L}_{GloU} + \mathcal{L}_{CE}) \end{aligned} \quad (8)$$

where $\mathcal{L}_{QFL}$, $\mathcal{L}_{L1}$, $\mathcal{L}_{GloU}$, $\mathcal{L}_{CE}$ and $\mathcal{L}_{Focal}$ are the quality focal loss, L1 loss, GloU loss, cross-entropy loss and focal loss. Hyperparameters λ s are the weighting factor for each loss term.

4. Experiments

4.1. Dataset and evaluation metric

FLIR [42]: It contains 5142 RGB-T image pairs captured during both daytime and nighttime. Due to the misalignment in the original dataset, the aligned version [43] is commonly chosen for the experiments. The dataset is divided into 4129 pairs for training and 1013 pairs for testing.

LLVIP [44]: It contains street scenes rich in pedestrians and cyclists, amounting to 15,488 RGB-T image pairs [44]. Following [44], 12,025 pairs are used for training and 3463 pairs for testing. Thermal images are primarily used for labeling, which is copied directly to RGB images.

M³FD [45]: It includes 4200 RGB-T aligned image pairs collected under various conditions such as different lighting, seasons, and weather scenarios. It covers six typical categories of automated driving and road surveillance, which is divided into training and testing sets with a ratio of 8:2 as provided in [46].

VEDAI [47]: It is designed for multi-class vehicle detection in aerial images, comprising 3640 objects instances across nine categories, including boats, cars, campers, airplanes, shuttles, tractors, trucks, vans, and others. It contains a total of 1210 images with a resolution of $1024 \times 1,024$, each featuring three RGB color channels and one NIR channel. The dataset is divided into training and test sets in a 9:1 ratio. Annotations, originally in a rotated box format with quadratic coordinates, are converted to a horizontal-box format following the methodology described in [48].

Average Precision (AP): The AP metric is derived from the area under the Precision-Recall curve, which plots recall on the horizontal axis and precision on the vertical axis. The mean Average Precision (mAP) is calculated by taking the weighted average of AP across all classes. In our experiments, we use an Intersection over Union (IoU) threshold of 0.5 to compute the mAP. Higher values of this metric indicate better performance.

Table 2

Comparison on the FLIR-align dataset. '-' indicates missing values. '‡' symbol denotes experimental results obtained using the CoDetr framework, with input images resized to a fixed resolution of 640×640 pixels).

Methods	mAP@0.5	mAP	Bicycle	Car	Person
MMTOD-CG [49]	61.4	-	50.3	70.6	63.3
MMTOD-UNIT [49]	61.5	-	49.4	70.7	64.5
CMPD [50]	69.4	-	59.9	78.1	69.6
CFR [26]	72.4	-	57.8	84.9	74.5
GAFF [27]	72.9	37.5	-	-	-
BU-ATT [51]	73.1	-	56.1	87.0	76.1
BU-LTT [51]	73.2	-	57.4	86.5	75.6
UA_CMDet [52]	78.6	-	64.3	88.4	83.2
CFT [2]	78.7	40.2	-	-	-
CSAA [53]	79.2	41.3	-	-	-
ICAFusion [8]	79.2	41.4	66.9	89.0	81.6
CrossFormer [54]	79.3	42.1	-	-	-
MFPT [55]	80.0	-	67.7	89.0	83.2
MMFN [34]	80.8	41.7	65.5	91.2	85.7
RSdet [56]	81.1	41.4	-	-	-
MiPa [57]	81.3	44.8	-	-	-
UniRGB-IR [58]	81.4	44.1	-	-	-
CPCF [32]	82.1	44.6	-	-	-
GM-DETR [59]	83.9	45.8	-	-	-
Fusion-Mamba(yolov5) [60]	84.3	44.4	-	-	-
Fusion-Mamba(yolov8) [60]	84.9	47.0	-	-	-
TFDet [33]	86.6	46.6	-	-	-
DAMSDet[30]	86.6	49.3	-	-	-
Ours	83.3	40.3	74.9	89.8	85.1
Ours‡	87.8	49.7	79.6	93.4	90.2

4.2. Experimental setup

All experiments are conducted using PyTorch on a computer equipped with an Intel i7-9700 CPU, 64 GB RAM, and a Nvidia RTX 3090 GPU with 24 GB of memory. For all ablation studies, the number of epochs is set to 60. In our experiments, the batch size is set to 4, and the SGD optimizer is used with an initial learning rate of 0.01 and a momentum of 0.937. The weight decay factor is set to 0.0005, and we employ a cosine learning rate decay schedule. The input size for all training images is 640×640 , while the input size for testing is 640×512 . Additionally, mosaic augmentation is used for data enhancement.

4.3. State-of-the-art comparison

The MS2Fusion module is experimented with both the YOLOv5 and CoDetr framework [41]. The YOLOv5 detector possesses faster inference speed but lower accuracy, while the CoDetr has better detection accuracy but slower inference speed.

4.3.1. Comparison on the FLIR dataset

Table 2 provides a comparative analysis of our method against existing approaches on the FLIR-align dataset. The results demonstrate that our method achieves the highest mAP@0.5 score of 83.3% in YOLOv5 framework and 87.8% in CoDetr framework across all classes, marking a significant improvement of 2.2% (YOLOv5) and 6.7% (CoDetr) over current state-of-the-art methods. Notably, the performance gain is particularly pronounced in the 'person' and 'bicycle' classes, highlighting our fusion method's superior effectiveness in addressing challenges related to non-thermal and small objects.

4.3.2. Comparison on the LLVIP dataset

Table 3 shows the performance metrics of our model on the LLVIP dataset, where our approach achieves superior results in both mAP@0.5 and mAP compared to existing models. Specifically, our method can obtain 97.7%(YOLOv5) and 98.4%(CoDetr) in terms of mAP@0.5, outperforming conventional CNN and Transformer-based approaches. This result underscores the effectiveness of our model in achieving state-of-the-art performance on the LLVIP dataset.

Table 3
Comparison on the LLVIP dataset.

Methods	mAP@0.5	mAP
DIVFusion [61]	89.8	52.0
GAFF [27]	94.0	55.8
CSAA [53]	94.3	41.3
ECISNet [62]	95.7	-
RSDet [56]	95.8	61.3
UniRGB-IR [58]	96.1	63.2
UA_CMDet [52]	96.3	-
CPCF [32]	96.4	65.0
Fusion-Mamba(yolov5) [60]	96.8	62.8
Fusion-Mamba(yolov8) [60]	97.0	64.3
MMFN [34]	97.2	-
GM-DETR [59]	97.4	70.2
TFDet [33]	97.9	71.1
DAMSDet[30]	97.9	69.6
MiPa [57]	98.2	66.5
Ours	97.5	65.5
Ours [‡]	98.4	70.6

4.3.3. comparison on the m³FD dataset

Table 4 provides a comparative analysis of our model against existing methods on the M³FD dataset. Our approach shows superior performance with a notable 3.2%(YOLOv5) and 5.2%(CoDet) improvement over other models. This improvement is particularly significant for high heat source objects such as 'motorcycles', showcasing the efficacy of our fusion approach in effectively integrating features from thermal images.

4.3.4. Comparison on the VEDAI dataset.

Table 5 highlights that our method achieves state-of-the-art performance on the VEDAI dataset, even without specialized techniques for detecting small objects. This underscores our model's strength in focusing on the multi-modal features of small objects, proving its effectiveness. These results indicate that our approach excels in UAV image applications, ensuring robust performance in detecting small objects.

4.4. Generalization to other multimodal tasks

4.4.1. Experiments on RGB-T semantic segmentation

Evaluation metrics: Mean Intersection over Union (mIoU) is a commonly used evaluation metric for semantic segmentation models, which measures the model performance by calculating the ratio between the intersection and union of predictions and ground truth segments.

MFNet dataset [72]: It is the first publicly available RGB-T dataset featuring pixel-level annotations. It consists of 1569 aligned RGB-T image pairs captured in urban environments, with semantic labels for eight common driving-scene obstacles: cars, pedestrians, bicycles, curves, bus stops, guardrails, traffic cones, and speed bumps.

SemanticRT [73] dataset: It consists of 11,371 high-quality, pixel-level annotated RGB-T image pairs. It is seven times larger than the existing MFNet dataset and covers a wide range of challenging scenes under unfavorable lighting conditions such as low light and pitch black.

Comparison on the MFNet dataset. As illustrated in Table 6, our MS2Fusion method performs best in this dataset, significantly improving mIoU by 2.8% compared to the baseline [72] (without any specialized design). The improvement is particularly noticeable on the 'Bump' and 'Curve' categories. Our analysis reveals that the "Bump" and "Curve" categories primarily rely on local geometric features of object surfaces, which constitute the shared features across multi-modal data. MS2Fusion can effectively leverage these shared features to significantly improve segmentation performance for these two categories.

Comparison on the SemanticRT dataset. As shown in Table 7, our MS2Fusion method also exhibits superior performance. Specifically, it achieves an improvement of 1.0% over the baseline and differs by only 0.5% from the current state-of-the-art model. These findings highlight the robustness and versatility of our approach in multispectral feature fusion, proving its adaptability to various tasks.

Based on the experimental comparisons on the two RGB-T semantic segmentation benchmarks, we demonstrate that our MS2Fusion is also equally effective, with a high degree of generality and adaptability to a wide range of downstream tasks.

4.4.2. Experiments on RGB-T salient object detection (RGB-T SOD)

Evaluation metrics: Following [97], Mean Absolute Error (MAE), F-Measure (adpF), S-Measure (S), and E-Measure (adpE) are commonly used as evaluation metrics for SOD.

VT821 dataset [82]: It provides 821 aligned RGB-T image pairs capturing challenging real-world scenarios. It specifically includes illumination variations, object occlusions, and low-contrast thermal conditions to test model robustness in dynamic settings.

VT1000 dataset [84]: It comprises 1000 pairs of RGB-T images, offering a broader range of scenes, including urban, rural, indoor, and outdoor environments. This dataset enhances diversity by covering different weather conditions, times of day (daylight and nighttime), and various object types (e.g., pedestrians, vehicles, animals).

VT5000 dataset [94]: It is a large-scale dataset with 5000 pairs of RGB-T images. It features a wide array of challenging conditions, such as extreme weather, dense occlusion, and multi-object scenes, offering high diversity and difficulty.

MSEDNET [6] is employed as our baseline RGB-T SOD method due to its proven hierarchical fusion architecture and superior performance, and MS2Fusion-MSEDNET denotes the baseline equipped with our proposed MS2Fusion module. As shown in Table 8, the MS2Fusion-MSEDNET method achieves state-of-the-art performance for RGB-T SOD on the VT821, VT1000 and VT5000 datasets as well. It is clear to observe that MS2Fusion-MSEDNET ranks first or second on all evaluation metrics, demonstrating its superior capabilities of feature fusion in the RGB-T SOD task.

4.5. Ablation studies

All ablation studies in this paper are conducted on the FLIR dataset. Unless otherwise specified, the experimental setting, model architecture, and parameter settings follow those described in Section 4.2.

4.5.1. Different detection frameworks

To thoroughly evaluate the generalization of the MS2Fusion module, comparative experiments are conducted on two mainstream detection frameworks, YOLOv5 and CoDet [41], with add fusion serving as the baseline method. Table 9 unequivocally demonstrate that our fusion approach achieves remarkable performance improvements in both frameworks, delivering a 3.3% boost in mAP@0.5 over the baseline. These findings strongly validate the efficacy and versatility of MS2Fusion. Notably, the module exhibits excellent plug-and-play characteristics, enabling seamless integration with various detection frameworks and providing a flexible yet powerful solution for multimodal object detection tasks.

4.5.2. Different backbones

To comprehensively evaluate the architectural compatibility of the MS2Fusion module, we conducted systematic experiments with three distinct backbone networks: VGG16 [98], ResNet50 [99], and CSPDarkNet53 [18]. As demonstrated in Table 10, MS2Fusion achieves consistent performance gains across all architectural configurations. Specifically, our method yields improvements of 1.0%, 4.8%, and 3.3% over the baseline when employing VGG16, ResNet50, and CSPDarkNet53 backbones, respectively. These experimental results not only validate the effectiveness of the MS2Fusion module but also substantiate its remarkable generalization capability across diverse backbone architectures.

Table 4
Comparison on the M³FD dataset.

Methods	mAP@0.5	mAP	People	Bus	Car	Motorcycle	Lamp	Truck
DIDFuse [63]	79.0	52.6	79.6	79.7	92.5	68.7	84.7	68.8
SDNet [64]	79.0	52.9	79.4	81.4	92.3	67.4	84.1	69.3
RFNet [65]	79.4	53.2	79.4	78.2	91.1	72.8	85.0	69.0
ReC [66,67]	79.5	-	79.4	78.9	91.8	69.3	87.4	70.0
U2F [67,68]	79.6	-	80.7	79.2	92.3	66.8	87.6	71.4
DAMSDet [30]	80.2	52.9	-	-	-	-	-	-
TarDAL [45]	80.5	54.1	81.5	81.3	94.8	69.3	87.1	68.7
DeFusion [69]	80.8	53.8	80.8	83.0	92.5	69.4	87.8	71.4
CDDFusion [67]	81.1	54.3	81.6	82.6	92.5	71.6	86.9	71.5
IGNet [70]	81.5	54.5	81.6	82.4	92.8	73.0	86.9	72.1
SuperFusion [71]	83.5	56.0	83.7	93.2	91.0	77.4	70.0	85.8
Fusion-Mamba(yolov5) [60]	85.0	57.5	80.3	92.8	91.9	73.0	84.8	87.1
MMFN [34]	86.2	-	83.0	92.1	93.2	73.7	87.6	87.4
Fusion-Mamba(yolov8) [60]	88.0	61.9	84.3	94.2	92.9	80.5	87.5	88.8
Ours	89.4	59.7	85.6	93.7	93.9	82.4	90.8	89.9
Ours‡	91.4	65.6	89.8	95.0	94.7	88.2	88.3	92.4

Table 5
Comparison on the VEDAI dataset.

Methods	mAP@0.5	Car	Truck	Pickup	Tractor	Camper	Ship	Van	Plane
ICAFusion [8]	76.6	88.0	67.5	80.9	74.9	78.3	56.9	69.8	96.7
MidFusion1 [48]	77.4	88.2	84.9	81.7	68.4	72.2	71.0	53.6	99.5
MidFusion2 [48]	78.9	89.2	78.8	87.6	66.2	73.2	62.5	73.9	99.5
Input Fusion1 [48]	80.0	88.7	77.7	79.5	72.3	75.7	78.6	72.1	95.7
Input Fusion2 [48]	80.5	89.1	84.9	78.7	84.2	70.1	69.9	68.8	98.4
YOLOFusion [48]	81.6	91.7	78.1	85.9	71.9	78.9	71.7	75.2	99.5
Ours	80.2	89.5	75.6	84.2	81.6	75.9	61.8	76.4	97.0
Ours‡	84.2	93.3	78.9	88.5	85.8	80.6	65.5	83.6	97.1

Table 6
Comparison on the MFNet dataset.

Methods	mIoU	Car	Person	Bike	Curve	Car Stop	Guardrail	Color Cone	Bump
PSTNet [74]	48.4	76.8	52.6	55.3	29.6	25.1	15.1	39.4	45.0
RTFNet [75]	53.2	87.4	70.3	62.7	45.3	29.8	0.0	29.1	55.7
FuseSeg [76]	54.5	87.9	71.7	64.6	44.8	22.7	6.4	46.9	47.9
AFNet [77]	54.6	86.0	67.4	62.0	43.0	28.9	4.6	44.9	56.6
ABMDRNet [78]	54.8	84.8	69.6	60.3	45.1	33.1	5.1	47.4	50.0
FEANet [79]	55.3	87.8	71.1	61.1	46.5	22.1	6.6	55.3	48.9
GMNet [80]	57.3	86.5	73.1	61.7	44.0	42.3	14.5	48.7	47.4
EGFNet [7]	57.5	89.8	71.6	63.9	46.7	31.3	6.7	52.0	57.4
DPLNet [81]	59.3	-	-	-	-	-	-	-	-
CMX [28]	59.7	90.1	75.2	64.5	50.2	35.3	8.5	54.2	60.6
CRM-RGBT-Seg [5]	61.4	90.0	75.1	67.0	45.2	49.7	18.4	54.2	54.4
MFNet(baseline) [72]	63.5	92.6	82.1	78.2	89.6	24.1	1.2	46.2	94.2
MS2Fusion-MFNet	66.3	94.4	82.5	81.0	89.9	34.7	0.0	49.7	98.3

Table 7
Comparison on the SemanticRT dataset.

Methods	mIoU	CarStop	Bike	Bicyclist	Mtcycle	Mtcyclist	Car	Tricycle	TrafficLight	Box	Pole	Curve	Person
PSTNet [74]	68.0	71.1	62.3	58.5	47.3	55.2	85.4	44.2	75.7	83.0	71.7	62.2	72.2
RTFNet [75]	75.5	79.6	68.0	67.4	63.7	61.6	90.4	66.0	78.3	85.9	78.0	67.2	78.9
EGFNet [7]	77.4	78.6	71.3	70.9	68.4	66.1	90.5	71.5	80.4	85.4	76.5	66.9	83.7
ECM [73]	79.3	80.2	75.0	75.5	71.4	70.4	90.3	74.0	85.9	85.6	77.2	68.3	85.0
MFNet(baseline) [72]	77.8	75.3	77.1	63.2	71.1	57.3	97.9	66	85.9	89.5	82.4	85.0	83.2
MS2Fusion-MFNet	78.8	75.3	77.7	66.3	72.3	59.4	97.9	68.2	86.0	89.6	81.8	84.6	86.7

4.5.3. Different modules

As shown in Table 11, our comparative study of CP-SSM, SP-SSM, and FF-SSM modules reveals their complementary strengths in object detection. CP-SSM excels in enhancing modality-specific features, improving mAP by 2.1 % and boosting challenging category detection (e.g., bicycles) by 5.4 %. SP-SSM focuses on extracting cross-modal shared features with strong generalization, while FF-SSM delivers efficient heterogeneous feature fusion, yielding a 2.5 % mAP gain with minimal overhead and consistent performance across categories.

Combining modules further amplifies performance. SP-SSM + FF-SSM achieves 82.9 % mAP, while integrating all three modules pushes mAP to 83.3 %, improving bicycle detection by 7.6 %. This synergy balances feature complementarity, sharing, and fusion, enhancing accuracy while keeping parameter growth reasonable.

In summary, CP-SSM, SP-SSM, and FF-SSM work in concert to boost detection via modality-specific enhancement, cross-modal generalization, and adaptive fusion-preserving unique features, uncovering shared semantics, and reducing redundancy effectively.

Table 8Comparison on the VT821, VT1000 and VT5000 datasets. ($\downarrow$ indicates smaller value is better, while $\uparrow$ indicates larger value is better).

Methods	VT821				VT1000				VT5000			
	S $\uparrow$	adpE $\uparrow$	adpF $\uparrow$	MAE $\downarrow$	S $\uparrow$	adpE $\uparrow$	adpF $\uparrow$	MAE $\downarrow$	S $\uparrow$	adpE $\uparrow$	adpF $\uparrow$	MAE $\downarrow$
MTMR [82]	72.5	81.5	66.2	10.9	70.6	83.6	71.5	11.9	68.0	79.5	59.5	11.4
M3S-NIR [83]	72.3	85.9	76.4	14.0	72.6	82.7	71.7	14.5	65.2	78.0	57.5	16.8
SGDL [84]	76.5	84.7	76.1	8.5	78.7	85.6	76.4	9.0	75.0	82.4	67.2	8.9
PoolNet [85]	75.1	73.9	57.8	10.9	83.4	81.3	71.4	6.7	76.9	75.5	58.8	8.9
R ³ Net[86]	78.6	80.9	66.0	7.3	84.2	85.9	76.1	5.5	75.7	79.0	61.5	8.3
CPD [87]	82.7	83.7	71.0	5.7	90.6	90.2	83.4	3.2	84.8	86.7	74.1	5.0
MMCI [88]	76.3	78.4	61.8	8.7	88.6	89.2	80.3	3.9	82.7	85.9	71.4	5.5
AFNet [89]	77.8	81.6	66.1	6.9	88.8	91.2	83.8	3.3	83.4	87.7	75.0	5.0
TANet [90]	81.8	85.2	71.7	5.2	90.2	91.2	83.8	3.0	84.7	88.3	75.4	4.7
S2MA [91]	81.1	81.3	70.9	9.8	91.8	91.2	84.8	2.9	85.3	86.4	74.3	5.3
JLDCF [92]	83.9	83.0	72.6	7.6	91.2	89.9	82.9	3.0	86.1	86.0	73.9	5.0
FMCF [93]	76.0	79.6	64.0	8.0	87.3	89.9	82.3	3.7	81.4	86.4	73.4	5.5
ADF [94]	81.0	84.2	71.7	7.7	91.0	92.1	84.7	3.4	86.4	89.1	77.8	4.8
MIDD [3]	87.1	89.5	80.3	4.5	91.5	93.3	88.0	2.7	86.8	89.6	79.9	4.3
LSNet [95]	87.2	91.0	81.4	3.6	92.1	94.9	88.1	2.3	87.5	92.0	81.7	3.7
CAVER [6,96]	89.8	92.8	87.7	2.7	93.6	94.9	91.1	1.7	89.9	94.1	84.9	2.8
DPLNet [81]	87.8	90.8	81.0	4.3	92.8	95.1	88.1	2.2	87.9	91.6	82.8	3.8
MS2Fusion(baseline) [6]	87.6	89.7	80.3	3.9	92.8	93.9	86.8	2.2	88.1	91.6	82.1	3.7
MS2Fusion-MS2Fusion	90.4	93.5	86.3	3.2	94.4	97.2	91.8	1.6	90.2	94.2	86.4	3.0

Table 9

Effect of different backbones and detection frameworks.

Num	Framework	Fusion Model	Person	Car	Bicycle	mAP@0.5	FPS
1	YOLOv5	Baseline	83.8	88.9	67.3	80.0	43.2
2		MS2Fusion	85.1	89.8	74.9	83.3 (+3.3)	24.4
3	CoDet	Baseline	88.0	91.7	73.8	84.5	5.9
4		MS2Fusion	90.2	93.4	79.6	87.8 (+3.3)	4.8

Table 10

Effect of different backbones.

Num	Backbone	Fusion Model	Person	Car	Bicycle	mAP@0.5
1	VGG16	Baseline	78.9	87.7	51.2	72.6
2		MS2Fusion	79.0	87.8	53.8	73.6 (+1.0)
3	ResNet50	Baseline	76.8	85.7	44.6	69.0
4		MS2Fusion	80.2	88.4	52.8	73.8 (+4.8)
5	CSPDarkNet53	Baseline	83.8	88.9	67.3	80.0
6		MS2Fusion	85.1	89.8	74.9	83.3 (+3.3)

Table 11

Effect of different fusion modules.

Num	CP-SSM	SP-SSM	FF-SSM	Person	Car	Bicycle	mAP@0.5	Params(M)
1				83.8	88.9	67.3	80.0	72.7
2	✓			83.8	89.8	72.7	82.1 (+2.1)	84.5
3		✓		85.6	90.4	70.6	82.2 (+2.2)	90.4
4			✓	84.9	90.1	72.4	82.5 (+2.5)	86.0
5	✓	✓		84.9	90.3	72.9	82.7 (+2.7)	117.0
6	✓		✓	85.5	90.0	72.5	82.7 (+2.7)	97.8
7		✓	✓	85.5	90.3	72.9	82.9 (+2.9)	103.7
8	✓	✓	✓	85.1	89.8	74.9	83.3 (+3.3)	130.2

4.5.4. Different fusion layers

As shown in Table 12, the incremental addition of MS2Fusion modules across different feature stages yields distinct performance characteristics. Initial test at only the P3 stage (Row 2) demonstrates a trade-off effect, where 'bicycle' AP experiences a substantial 5.7% improvement to 71.4% at the cost of marginal decreases in 'person' (83.4%) and 'car' (88.5%) detection, ultimately elevating the mAP to 81.1%. Expanding the fusion to both P3 and P4 stages (Row 3) reverses this pattern, boosting 'person' and 'car' AP to 85.8% and 90.1%. When employing fusion at all three stages (Row 4), it achieves optimal balance. While maintaining strong performance on 'person' (85.1%) and 'car' (89.8%), it delivers a remarkable 74.9% AP for 'bicycles', the highest among all

configurations and pushes the overall mAP to 83.3%. This systematic evaluation clearly demonstrates that multi-level feature fusion enables more effective cross-modal feature integration, with full-stage implementation proving particularly advantageous for challenging categories like 'bicycles' while preserving performance on other objects.

4.5.5. Discussion on CP-SSM

Different expanding methods. Following VMamba [35], we also explore various scanning orientations for SSM in Table 13. As shown in Table 13(row 1 ~ 3), we find that multidirectional scanning adversely impacts experimental outcomes, rendering it unsuitable for object detection tasks. We think that multidirectional scanning may alter object

Table 12
Effect of fusion at different layers.

Number	P3	P4	P5	Person	Car	Bicycle	mAP@0.5
1				83.8	88.9	67.3	80.0
2	✓			83.4	88.5	71.4	81.1 (+1.1)
3	✓	✓		85.8	90.1	70.0	82.0 (+2.0)
4	✓	✓	✓	85.1	89.8	74.9	83.3 (+3.3)

Table 13
Finetunes in CP-SSM.

Num	Finetune	Person	Car	Bicycle	mAP@0.5
1	rows	85.1	89.8	74.9	83.3
2	columns	64.1	90.3	72.8	82.4
3	rows and columns	85.4	90.4	70.2	82.0
4	w/o exchange C	85.7	90.6	71.7	82.7
5	exchange C	85.1	89.8	74.9	83.3

Table 14

Performance of different inputs. (V, T denotes Visible and thermal, respectively. V + T represents the input with dual modalities, while V + V or T + T denotes input with a single modality).

Num	Model	Input	mAP@0.5
1	YOLOv5	V	67.8
2		T	73.9
3	Baseline	V + V	61.2
4		T + T	77.8
5		V + T	80.0
6	Ours	V + V	68.4 (+7.2)
7		T + T	82.1 (+4.3)
8		V + T	83.3 (+3.3)

features, contradicting the stability required for accurate object detection.

Exchange C -parameters. As shown in Table 13, the experimental study of the cross-branch C -parameter exchange mechanism in the CP-SSM module demonstrates that exchanging hidden state projection matrices can significantly improve model performance. Compared to non-exchange configurations, this mechanism achieves a stable performance improvement of 0.6 %, not only enhancing the feature capture capability for complex data relationships but also optimizing cross-feature interaction efficiency. Experimental validation shows that this parameter exchange strategy is particularly effective in scenarios requiring fine-grained data representation, enabling the CP-SSM module to demonstrate outstanding performance in cross-modal feature extraction tasks. This finding highlights the importance of parameter interaction mechanism design in modern feature learning architectures.

4.5.6. Comparison with different input modalities

Table 14 evaluates the performance of our model under conditions when certain input modalities are missing. The results demonstrate that our model can achieve competitive results even when only one modality (V + V or T + T) is used. Specifically, when only thermal images are provided, our model's performance is just 1.2 % lower than that achieved with multi-modal inputs (as seen in Rows 7 and 8). This highlights the robustness of our approach in handling incomplete modality inputs. Additionally, our method shows substantial improvements over the baseline when the same two modalities are used (comparing Rows 3 and 6, 4 and 7, 5 and 8). This improvement underscores the effectiveness of our module in exploiting the shared spatial features of heterogeneous data. By enhancing the generalization of cross-modal features, our model effectively utilizes information from one available modality to enhance another modality's features. These findings indicate that our approach

Table 15
Input Configuration for FF-SSM Module.

FF-SSM input	mAP@0.5	person	car	bicycle
(V, T)	83.1	85.4	90.2	73.9
(T, V)	82.5	85.1	90.1	72.4
((V, T), (T, V))	83.3	85.1	89.8	74.9

Table 16

Statistical analysis of the model(Perform the experiment 10 times with random seeds released.).

	Mean(%) mAP@0.5	VAR(%) mAP	mAP@0.5	mAP
FLIR	82.99	40.60	0.15	2.47
LLVIP	97.41	63.46	0.03	2.31
M ³ FD	89.11	59.76	0.18	0.42

not only maintains high performance with reduced input data but also significantly leverages the shared and complementary features across different modalities.

4.5.7. Visualization of RGB-T semantic segmentation samples

4.5.8. Analysis of FF-SSM input configurations

In this section, we systematically evaluate the FF-SSM module with three distinct input configurations: (V, T), (T, V), and bidirectional ((V, T), (T, V)). As shown in Table 15, single-order inputs exhibit significant performance variations: the (V, T) configuration achieves 83.1 % mAP@0.5 (0.2 % decrease from baseline), with particularly pronounced accuracy degradation for 'bicycle' detection (1.0 % drop). The (T, V) configuration shows more substantial performance decline, reaching only 82.5 % mAP@0.5 (0.8 % decrease) and suffering a 2.5 % accuracy reduction for 'bicycle' detection. These results reveal a critical feature attenuation phenomenon, where early input features are gradually forgotten during state propagation. To overcome this limitation, we propose a novel bidirectional architecture that establishes robust cross-modal feature memory pathways. Our solution not only elevates overall performance to 83.3 % mAP@0.5 but also delivers significant improvements in 'bicycle' detection accuracy (74.9 %), while maintaining stable detection precision for both 'person' and 'car' categories.

As illustrated in Fig. 9, we conducted a comprehensive comparison of three ERF map configurations for the FF-SSM module: (V, T), (T, V), and bidirectional ((V, T), (T, V)). The results demonstrate that our bidirectional input strategy yields a substantially expanded effective receptive field compared to unidirectional approaches. This empirical evidence strongly corroborates the efficacy of our proposed bidirectional architecture in enhancing the model's receptive field capacity. The innovative design successfully mitigates feature attenuation in state-space models while maintaining linear computational complexity, thereby providing an efficient and effective solution for cross-modal feature fusion.

4.6. Qualitative analysis

4.6.1. Statistical analysis of the model

The model demonstrates excellent robustness and stability based on the analysis of 10 random seed experiments as shown in Table 16. On the FLIR dataset, it achieves an mAP@0.5 of 82.99 % with a variance of only 0.15 %. Its performance on the LLVIP dataset is particularly outstanding, reaching an mAP@0.5 of 97.41 % with an extremely low variance of 0.03 %. On the M³FD dataset, the model attains an mAP@0.5 of 89.11 % and an mAP variance of just 0.42 %. These results indicate that the model exhibits remarkable stability, fully demonstrating its strong resistance to variations in initial conditions. It maintains consistent performance across different data distributions, showcasing excellent practical engineering value.

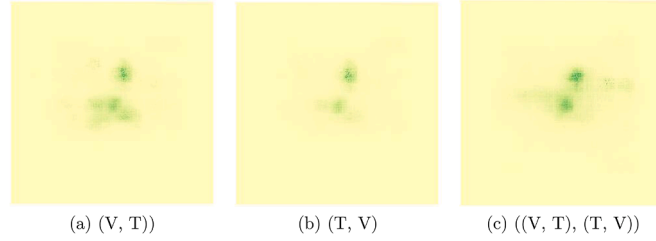


Fig. 9. ERF visualizations comparing different input configurations of the FF-SSM module: (a) unidirectional (V, T) input, (b) unidirectional (T, V) input, and (c) our proposed bidirectional ((V, T), (T, V)) input. The ERF map demonstrates that the bidirectional strategy achieves a significantly broader receptive field compared to the unidirectional approaches.

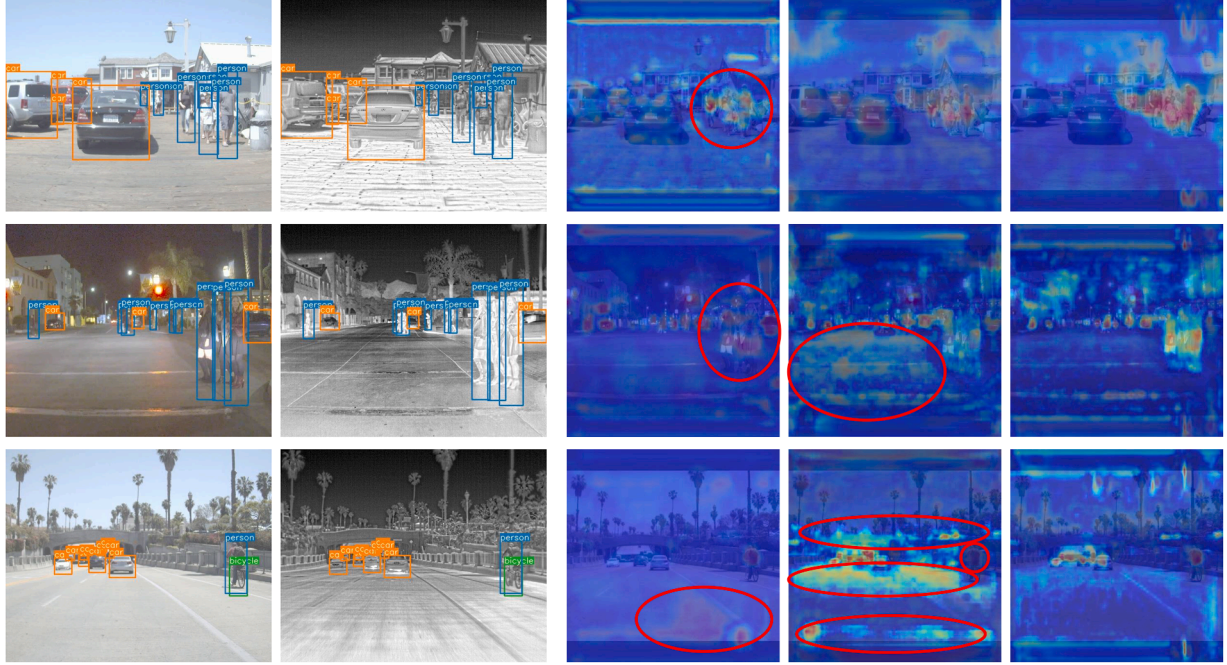


Fig. 10. Heatmap visualization. (The first and second columns are visible and thermal images; The third, fourth and fifth columns are heatmaps of baseline, ICAFusion and MS2Fusion, respectively).

4.6.2. Heatmap visualization

Fig. 10 presents the heatmap comparisons of three competing methods (baseline, ICAFusion and MS2Fusion) on RGB-T images. Through comprehensive analysis across multiple scenarios, the MS2Fusion method demonstrates consistent superiority. In the parking lot scenario (first row), the baseline method manages to detect partial instances of cars and pedestrians, yet produces significantly incomplete bounding boxes. Although ICAFusion shows noticeable improvement by capturing more objects, it still suffers from occasional missed detections. By contrast, MS2Fusion achieves near-perfect performance, precisely localizing all cars and pedestrians with highly accurate bounding boxes. The performance gap becomes even more pronounced in the challenging nighttime street scene (second row). While the baseline method can identify some pedestrians and vehicles, its detection accuracy proves inadequate. ICAFusion offers moderate improvement over the baseline, yet still struggles with low detection accuracy in these low-light conditions. Remarkably, MS2Fusion maintains excellent performance, reliably identifying all objects with exceptional precision even in this demanding scenario. Similarly, in the road scene (third row), the baseline method exhibits several limitations, including missed detections of vehicles and cyclists. ICAFusion partially addresses these issues by detecting more instances, but its accuracy remains suboptimal. Impressively, MS2Fusion again outperforms both competitors, achieving better detec-

tion performance with precise bounding boxes and minimal false negatives.

The MS2Fusion method demonstrates superior performance across all evaluated scenarios, achieving both high detection accuracy and precise object localization. Quantitative and qualitative analyses reveal three key advantages: (1) Compared to CNN-based baseline methods, MS2Fusion captures broader object regions through its expanded receptive field; (2) Relative to Transformer-based ICAFusion, it achieves more precise contour fitting; and (3) It maintains the lowest false negative rate, particularly in challenging low-light conditions. These improvements stem from MS2Fusion's effective cross-modal fusion mechanism, which optimally combines complementary information from RGB and thermal modalities. The heatmap visualizations further confirm that MS2Fusion's hybrid architecture successfully balances wide coverage (CNN advantage) and precise localization (Transformer strength), establishing it as an effective solution for multispectral object detection.

4.6.3. Visualization of feature fusion comparisons

We selected the P5 layer features for visualization, where F_{rgb_p5} , F_{v_p5} , F_{fused_p5} denote the RGB features, Thermal features, and fused features, respectively. Fig. 11 demonstrates the superior performance of MS2Fusion compared to the baseline approach, particularly in feature preservation and enhancement. The baseline method suffers

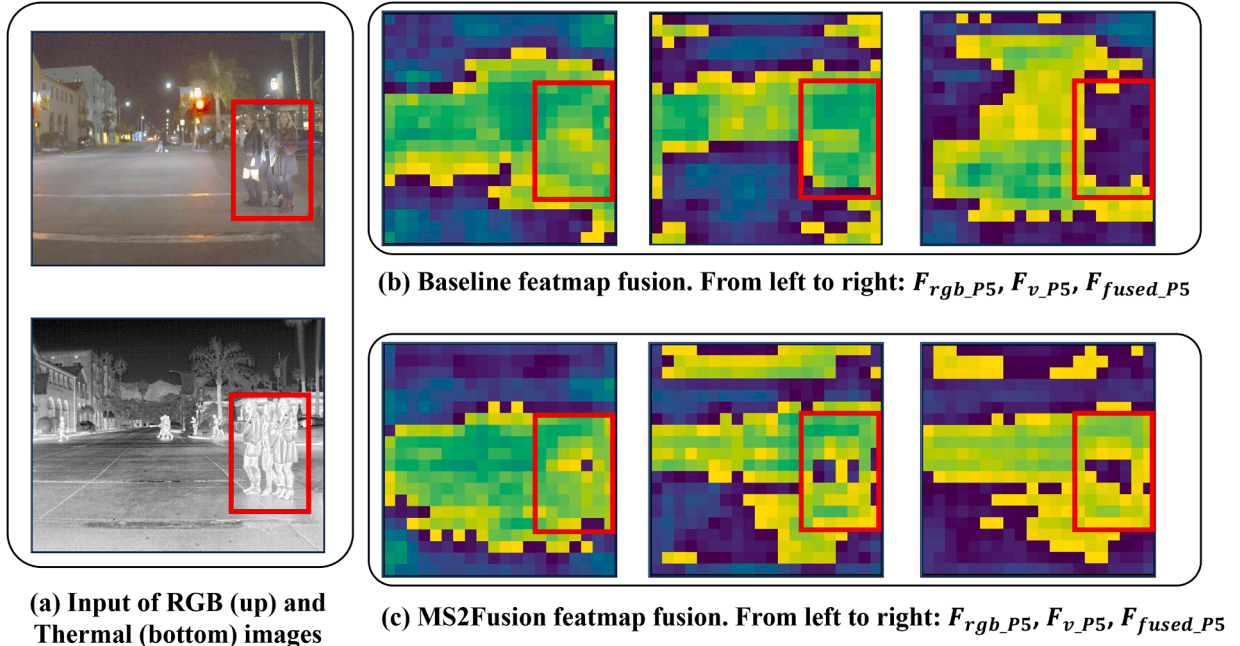


Fig. 11. Visualization of different feature map fusion.

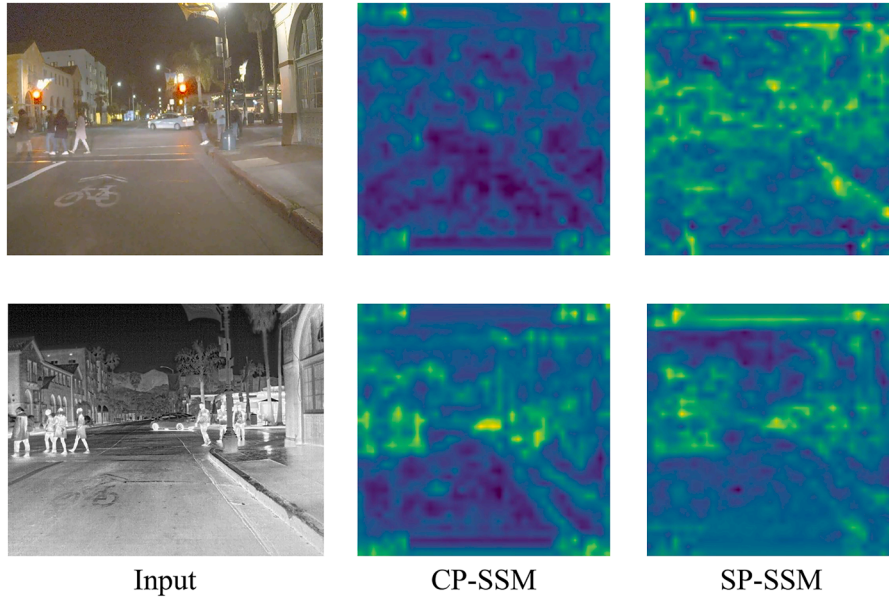


Fig. 12. Visualization of intermediate features under low-light conditions. The first column shows RGB and thermal inputs; the second and third columns display complementary (CP-SSM) and shared (SP-SSM) features, respectively.

from critical limitations: (1) it simply superimposes RGB and thermal feature maps through direct summation, often causing information loss or modal conflicts; (2) it fails to effectively leverage multimodal complementarity. These shortcomings are evident in its suboptimal feature representations.

In contrast, MS2Fusion addresses these issues through a sophisticated multi-stage fusion framework comprising three key modules. This hierarchical architecture enables selective retention of the most discriminative features from each modality while suppressing redundancy, as clearly visualized in the red rectangle of Fig. 11.

The framework's advantages manifest in enhanced feature retention that preserves modality-specific details, optimal complementarity utilization that dynamically balances RGB and thermal contributions, and improved scene adaptability that maintains robustness in complex en-

vironments. Quantitative results also confirm that MS2Fusion generates richer, clearer feature maps that directly translate to superior performance, resolving the fundamental trade-off between feature preservation and cross-modal integration that plagues the other methods.

4.6.4. Visual analysis of shared and complementary features.

As illustrated in Fig. 12, the first column shows the RGB and thermal input images, while the second and third columns visualize the complementary and shared feature maps learned by CP-SSM and SP-SSM, respectively. Due to the low-light conditions of this scene, the RGB modality captures relatively weak and sparse features, whereas the thermal modality provides richer structural and intensity cues. The CP-SSM focuses on these modality-specific differences, emphasizing distinctive thermal gradients and texture variations. In contrast, the SP-SSM

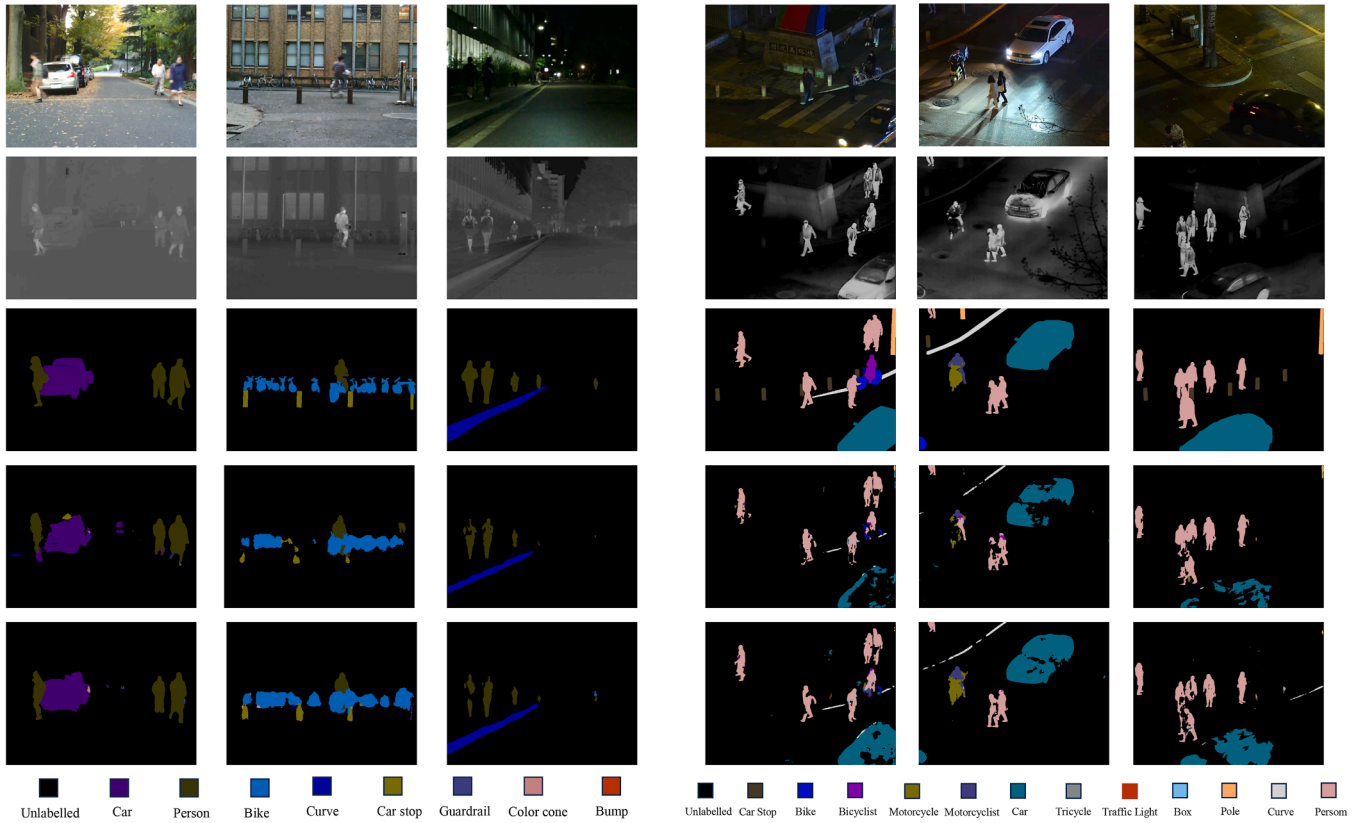


Fig. 13. Visualization of semantic segmentation of our model on MFNet and SemanticRT datasets. (The visible, thermal and groundtruth are provided in the first three rows, while the result of the baseline and MS2Fusion methods are shown in the fourth and fifth rows).

effectively fuses information from both modalities, producing shared semantic representations that consistently highlight object contours, pedestrian regions, and road boundaries across RGB and thermal domains. This indicates that SP-SSM compensates for illumination-induced degradation in the visible spectrum by enforcing semantic consistency and cross-modal alignment. Overall, this visualization demonstrates that MS2Fusion's dual-path design enables robust feature learning under challenging night-time conditions, balancing modality complementarity and shared semantic understanding.

Fig. 13 presents comparative semantic segmentation results across different input modalities and models. The visualization consists of five distinct rows: (1) visible input, (2) thermal input, (3) groundtruth annotations serving as reference labels, (4) baseline model predictions, and (5) MS2Fusion results. The baseline model exhibits noticeable deficiencies, particularly in handling pedestrian and vehicle boundaries, where segmentation appears blurred and misclassified. However, MS2Fusion demonstrates marked improvement in these challenging areas. The proposed method shows particular strength in preserving fine details along object contours and maintaining segmentation consistency in complex backgrounds, as evidenced by its precise delineation of persons and vehicles. Quantitative analysis confirms that MS2Fusion's multimodal fusion strategy effectively enhances segmentation accuracy by reducing classification errors and improving overall prediction quality compared to the baseline approach.

4.6.5. Visualization of RGB-T salient object detection samples

Fig. 14 presents a comprehensive comparison between our method (last column) and existing approaches for salient object detection. The visualization includes: (1) visible and thermal inputs (first two columns), (2) ground truth masks (third column), and (3) predictions from nine existing methods (remaining columns). Our method demonstrates su-

perior performance across multiple challenging scenarios through three key advantages: First, the proposed approach effectively leverages cross-modal complementarity between visible and thermal data to produce precise object boundaries with minimal noise interference (rows 1 and 5). Second, it exhibits remarkable robustness in occluded scenes (row 2), maintaining complete object morphology where competing methods generate fragmented or blurred detections. Third, our solution shows exceptional sensitivity to small objects (row 4) and complex shapes (rows 3 and 6), achieving detection accuracy that closely matches the ground truth.

Quantitative analysis reveals our method maintains consistent performance across diverse challenging conditions, including low-resolution inputs, occlusions, small objects, and cluttered backgrounds, significantly outperforming existing approaches that show scene-specific limitations. This demonstrates our framework's superior generalization capability and establishes it as a robust solution for salient object detection tasks.

4.7. Limitations

In this section, we analyze several failure cases that highlight the limitations of our method in Fig. 15:

- Distant objects: In the first row, the images illustrate scenarios where the object is distant and is not prominent in the thermal image. When the object's thermal signature closely matches the background, our method struggles to accurately distinguish the object from its surroundings.
- Occluded objects: The second row of images shows instances where objects are partially or fully occluded. Our method tends to overlook these occluded objects, leading to missed detections.

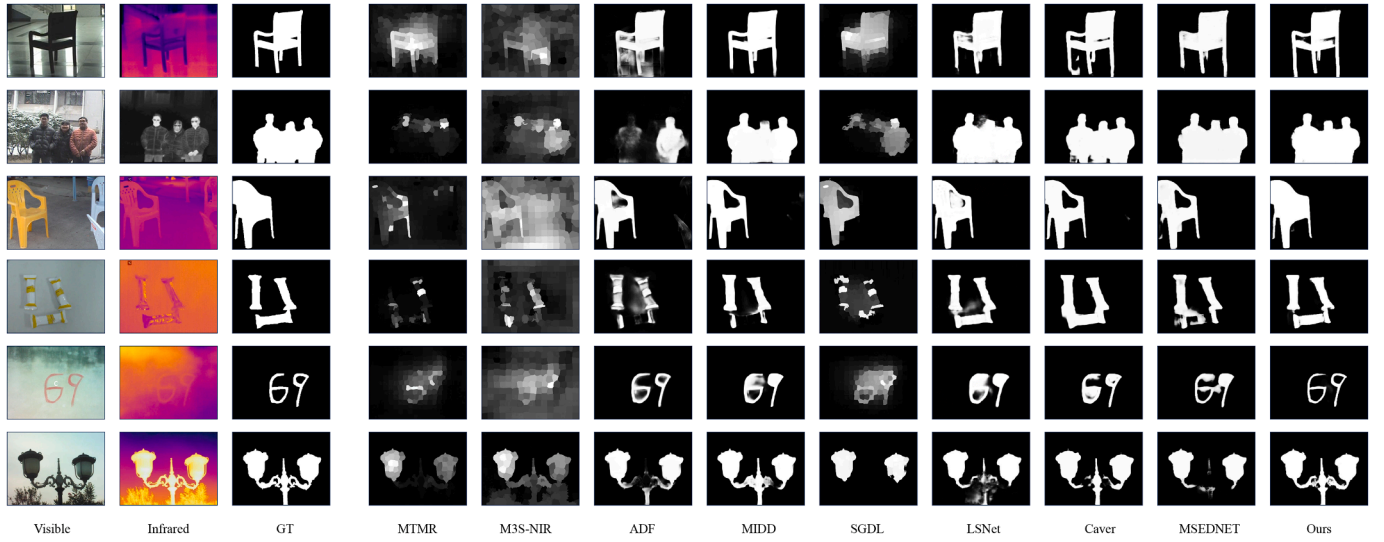


Fig. 14. Visual comparison among different SOTA methods and Ours.



Fig. 15. Failure cases on the FLIR dataset. (The first and second columns are the input RGB and thermal image, the third column is groundtruth, and the fourth column is the detection result of our method. The red circles indicate the false positives or false negatives in the images. Zoom in for more details).

- False positives with similar thermal appearance: The third row demonstrates situations where objects with thermal characteristics similar to those of a person result in false-positive detections. This challenge is exacerbated by the low resolution of the images, which hampers the model's ability to differentiate between actual objects of interest and irrelevant background features.

These failure cases demonstrate the critical need to improve our model's capability to handle three key challenges: distant small object detection, occluded object detection, and discrimination between thermally similar objects, especially in low-resolution images.

5. Conclusion

In this paper, we propose MS2Fusion, an adaptive and efficient multispectral feature fusion framework that addresses the key limitations of existing methods. Unlike conventional approaches that rely on rigid fusion strategies or suffer from computational inefficiency, our method

leverages dynamic state space models to efficiently capture long-range dependencies while maintaining low computational complexity, overcoming the drawbacks of both CNN-based and Transformer-based techniques. Furthermore, we introduce a shared-parameter state space module to explicitly model cross-modal feature interactions, enhancing discriminative power while mitigating information loss and modal discrepancies. Extensive experiments demonstrate that MS2Fusion achieves state-of-the-art performance, validating its effectiveness in robust multispectral object detection under challenging conditions.

In future work, we plan to adapt MS2Fusion for resource-constrained platforms such as wireless sensor networks and UAV-based systems. Leveraging recent progress in efficient data transmission [100] and lightweight real-time detection for drones [101], we aim to integrate TinyML-oriented optimizations, including model pruning, quantization, and state-space compression, to enhance real-time efficiency and energy awareness. Beyond the current scope, the proposed fusion methodology will be extended to other multimodal sensing tasks under computational

and bandwidth constraints, supported by our ongoing construction of a multispectral UAV-maritime dataset for large-scale validation.

CRedit authorship contribution statement

Jifeng Shen: Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Conceptualization; **Haibo Zhan:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation; **Shaohua Dong:** Writing – review & editing, Visualization, Software, Methodology, Data curation; **Xin Zuo:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization; **Wankou Yang:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization; **Haibin Ling:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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